The Portable Coach:

Which NFL Coaching Effects Transfer Across Rosters, and How Confidently

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1 Introduction

NFL franchises hire and fire coaches as though coaching effects are large, transferable, and known. Whether they are any of those things is an empirical question with nine-figure consequences: a coordinator hire is a bet that his scheme travels, and a head-coach hire is a bet that his identity does. We ask which coaching effects actually transfer to a new roster, at what resolution, and — the decision-relevant part — which are forecastable precisely enough to price.

2 Methods

We apply a single partition, the head-coach/offensive-coordinator/defensive-coordinator staff triple at each team-season, to nflverse play-by-play from 2002 to 2024, at four resolutions from macro tendencies to scheme-specific concepts, under one signal-to-noise discipline: a $B = 1000$ permutation null, cross-season holdout, and a propensity-matched event design for coaching changes. Portability is estimated with a hierarchical random-coefficient model: the slope by which a coach's prior-tenure profile reappears at his next team. The identical framework is run on the NBA as a transfer check.

3 Results

Offense is two-locus. Head coaches author macro philosophy — mid-season firings shift pass rate $1.36\times$ ($p = .002$) and off-season changes shift pace $1.17\times$ ($p = .021$) — while offensive coordinators author the schematic fine grain, led by red-zone shotgun rate at $1.34\times$ ($p = .0002$). Portability separates the loci: the offensive-coordinator slope is 0.469 (95% CI [0.32, 0.62]), robust to team-quality adjustment and broad, with six distinct concept families porting at slopes 0.41-0.50. The head-coach slope is 0.279 with a CI bracketing zero ($[-0.01, 0.56]$): head-coach portability is real but lives in which coach, not in a population constant. Under proper role isolation, defensive-coordinator changes move opponent macro tendencies acutely in the modern era (opponent plays $6.37\times$, $p = .007$). Applied to the NBA, the framework surfaces a different surviving locus —

head-coach perimeter defense — showing the method transfers while the sport-specific finding does not.

4 Conclusion

Exactly one coaching locus is forecastable precisely enough to price: an offensive coordinator's schematic identity, roughly half of which reappears on a new roster, concept by concept. A front office can therefore value an OC hire from his prior-tenure fingerprint with stated confidence, while a head-coach hire remains a bet on macro philosophy whose portability must be argued coach by coach, not assumed from the population. The framework also gives the league a detection-limits discipline: we report what the deployment cannot see — quiet schematic drift below event resolution — rather than hiding it, because a pricing tool is only trustworthy at its stated resolution.

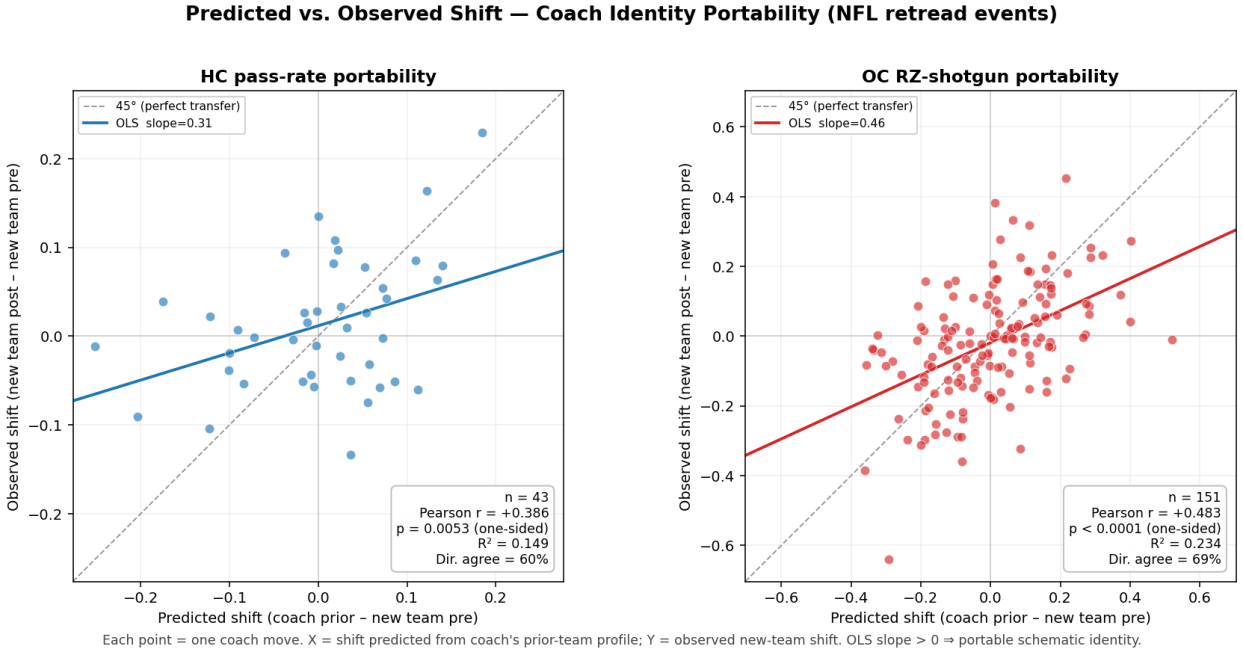


Figure 1: Predictive validation of coaching portability. Each point is a coach who changed teams; the horizontal axis is the schematic shift predicted from the coach's prior-tenure profile, the vertical axis the realized shift at the new team. The offensive-coordinator population slope (0.469, 95% CI [0.32, 0.62]) is the deployable portability constant; the head-coach slope (0.279, 95% CI [−0.01, 0.56]) brackets zero, indicating coach-level heterogeneity rather than a population mean shift.

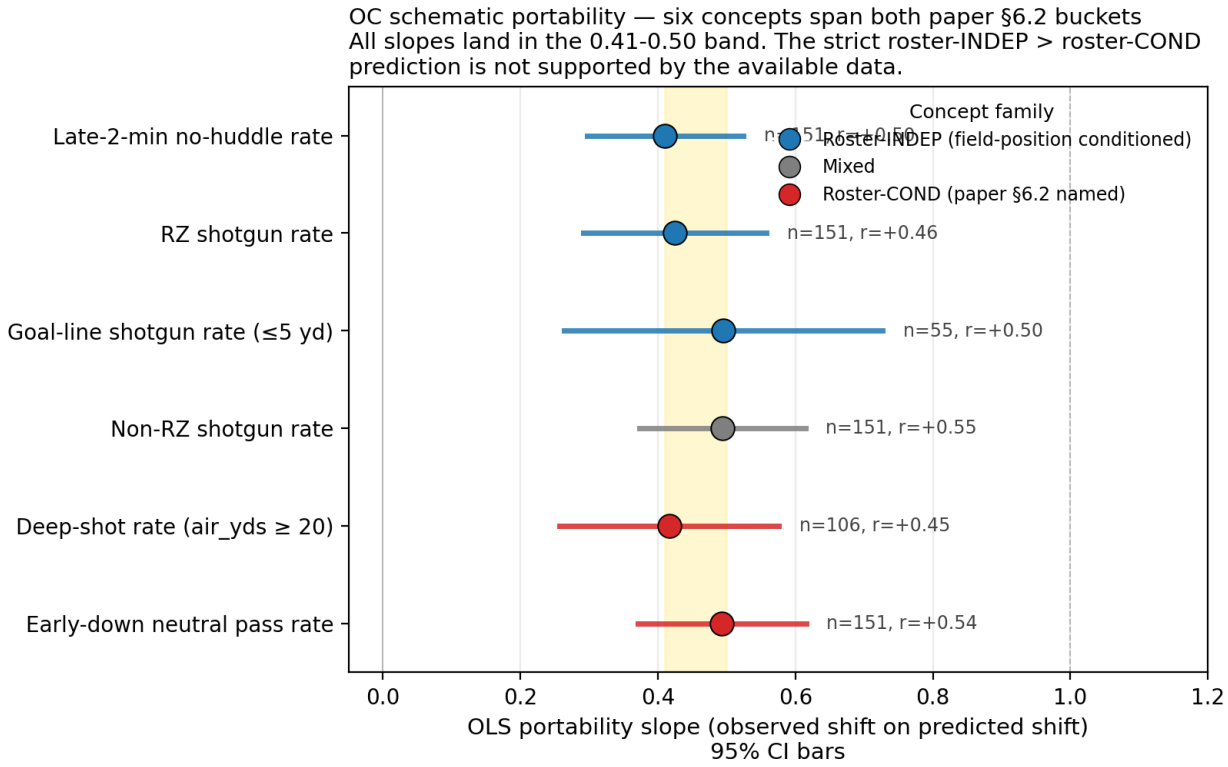


Figure 2: What a coordinator's whole schematic signature is worth, concept by concept. Each point is the predictive-validation slope (95% CI) at which a coordinator's prior-tenure rate in that concept reappears at a new team. Because all six transfer at comparable slopes (0.41–0.50), a front office can price a coordinator's full tendency profile, not just one statistic.